



BIOM1051

Module 1: Principles of Cell Function

Lectures 1 & 2

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Module 1: Principles of Cell Function

- Membrane Structure & Function – Lecture 1 & 2
- Cell Communication & Receptor Families – Lecture 3
- Review & Application – Workshop 2
- + the Osmosis Practical

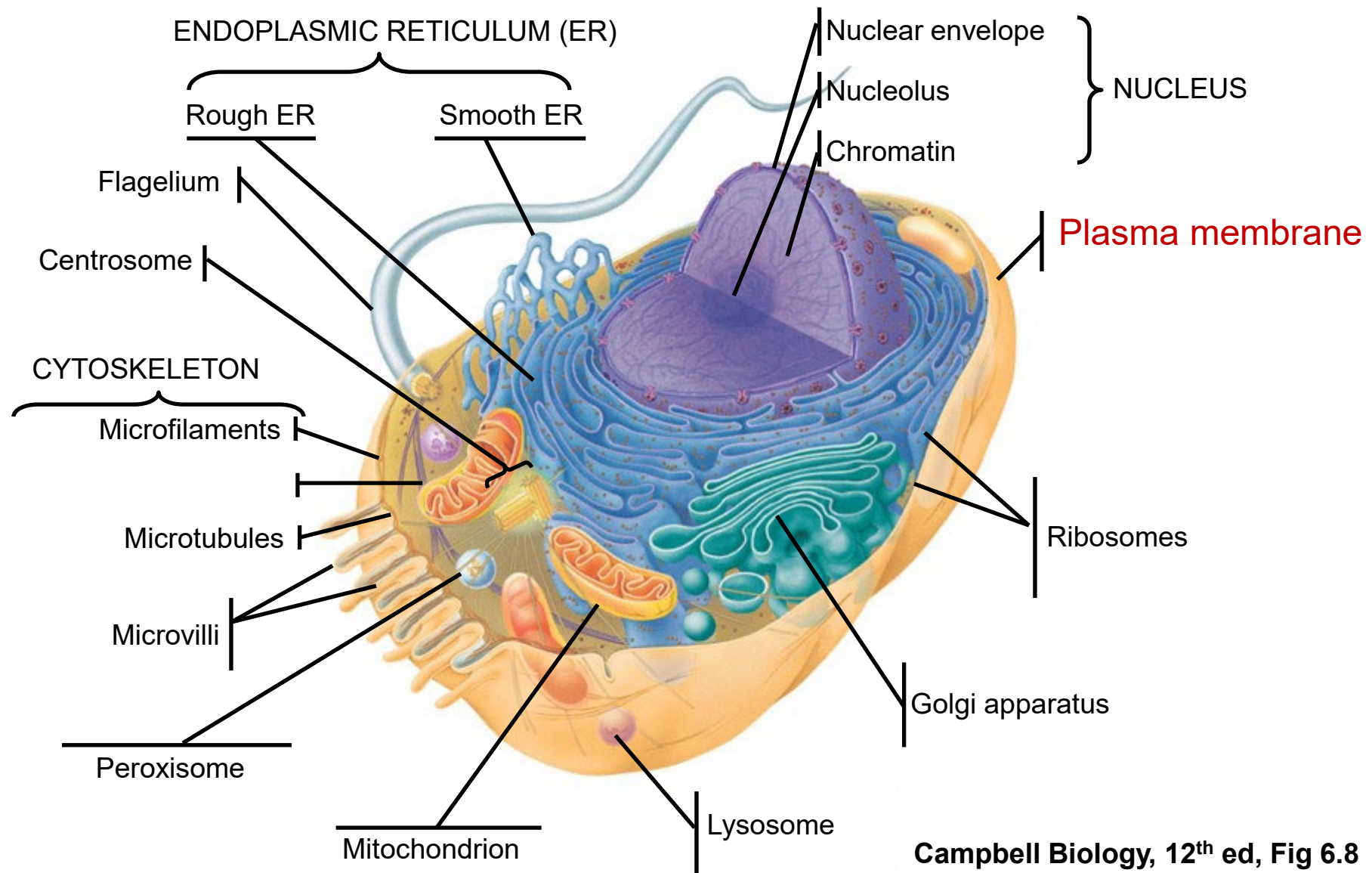
Membrane Structure & Function: Learning Objectives – Lectures 1& 2

- Describe the structural elements of cell membranes and their function
- List the properties of membrane proteins
- Describe how membrane structure results in selective permeability
- Rationalise the process of osmosis
- Compare and contrast passive diffusion, facilitated diffusion and active transport
- Describe the processes for bulk transport across the cell membrane

Membrane Structure & Function: Resources

- Campbell et al. *Biology*, 12th ed, 2021 – Chapter 7
 - Concept 7.1: Cellular membranes are fluid mosaics of lipids and proteins
 - Concept 7.2: Membrane structure results in selective permeability
 - Concept 7.3: Passive transport is diffusion of a substance across a membrane with no energy investment
 - Concept 7.4: Active transport uses energy to move solutes against their gradients
 - Concept 7.5: Bulk transport across the plasma membrane occurs by exocytosis and endocytosis
- Videos – in Bb folder (4 important topics)
- Mastering Biology – a wealth of learning aids

Recommended review reading: Campbell Biology, 12th ed, Chapter 6 – ‘A Tour of the Cell’
(Chapter 6 is for review and is not specifically examinable for BIOM1051).

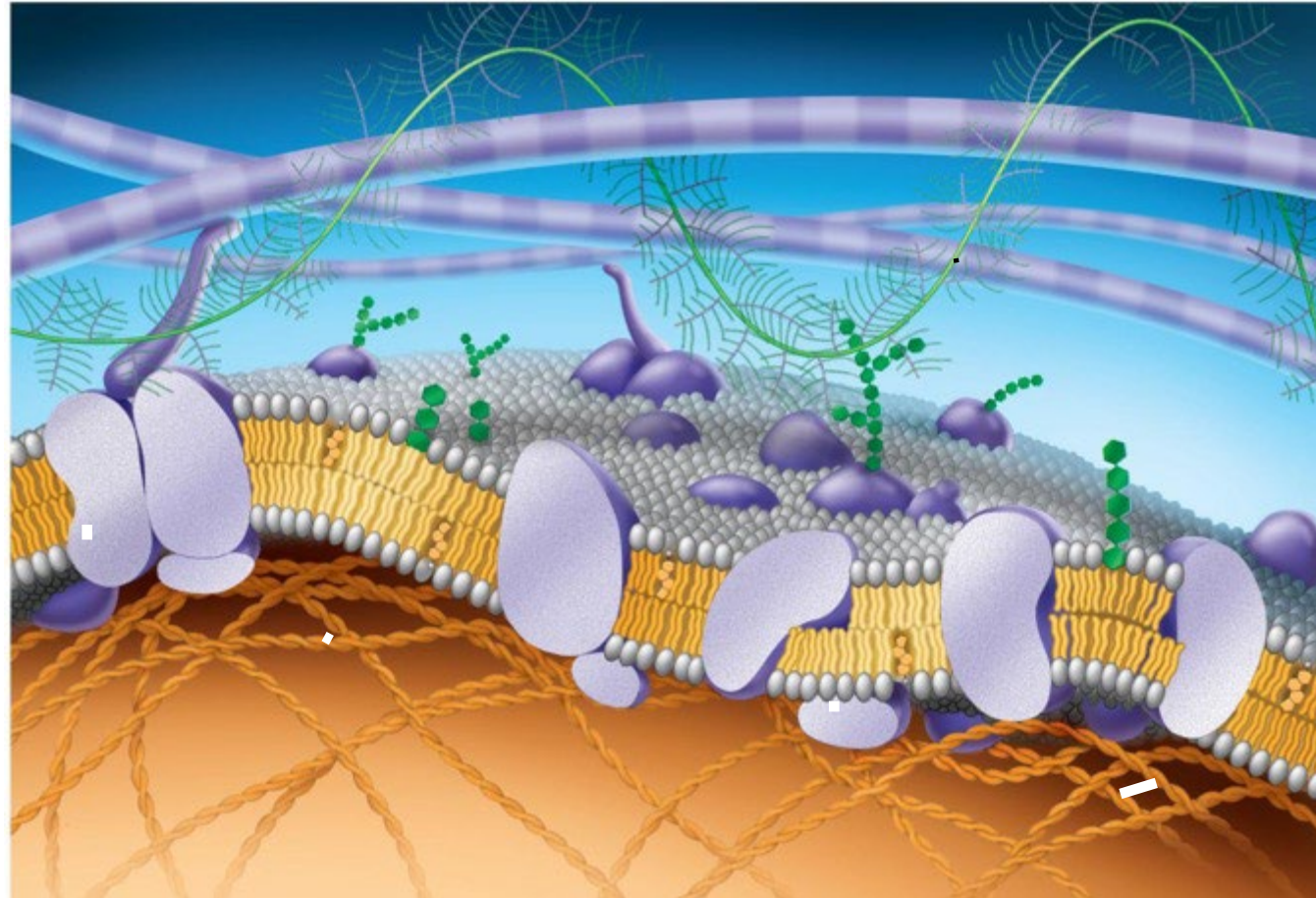


Campbell Biology, 12th ed, Fig 6.8

Concept 7.1: Cellular membranes are fluid mosaics of lipids and proteins

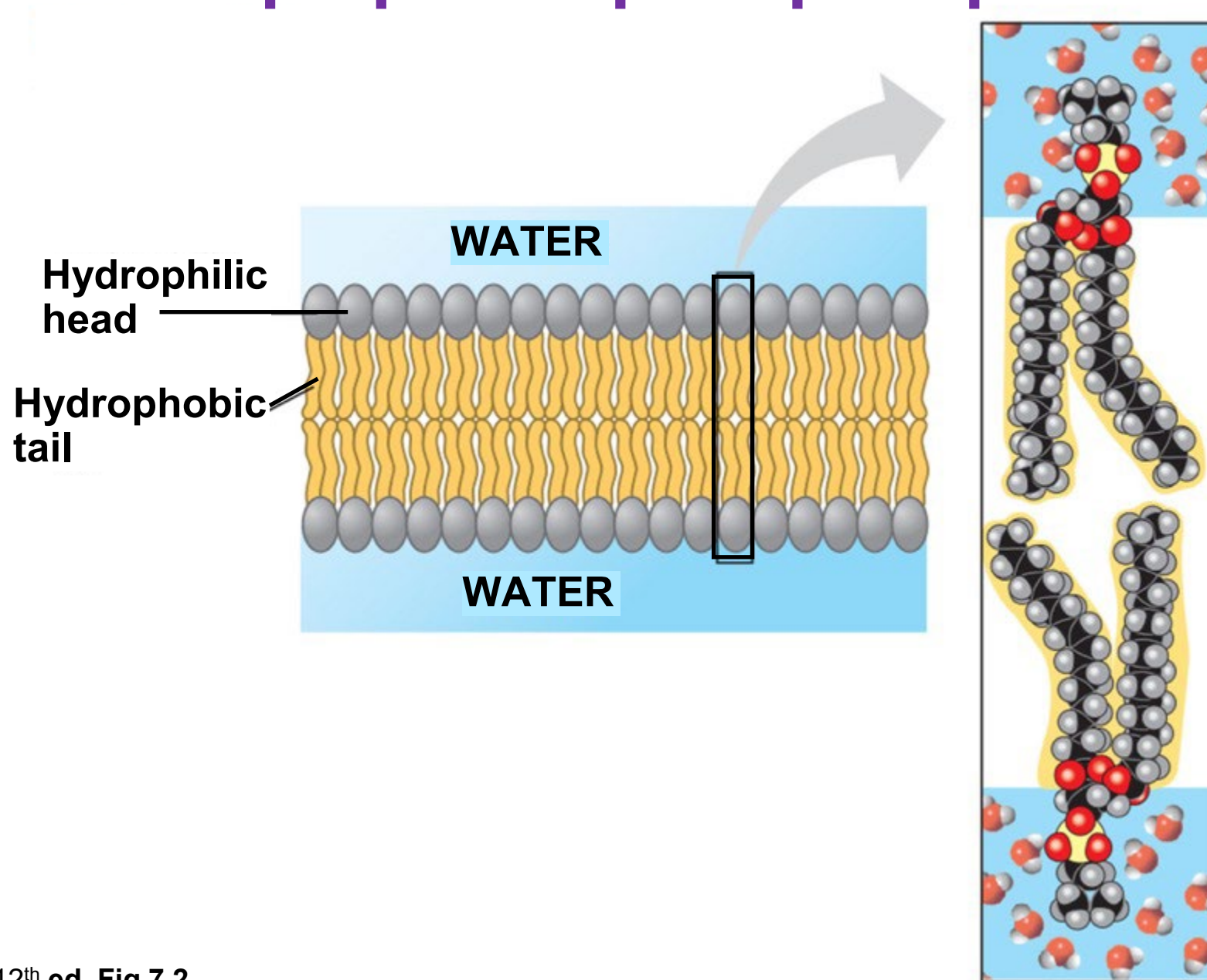
Phospholipids – abundant – **amphipathic molecules**

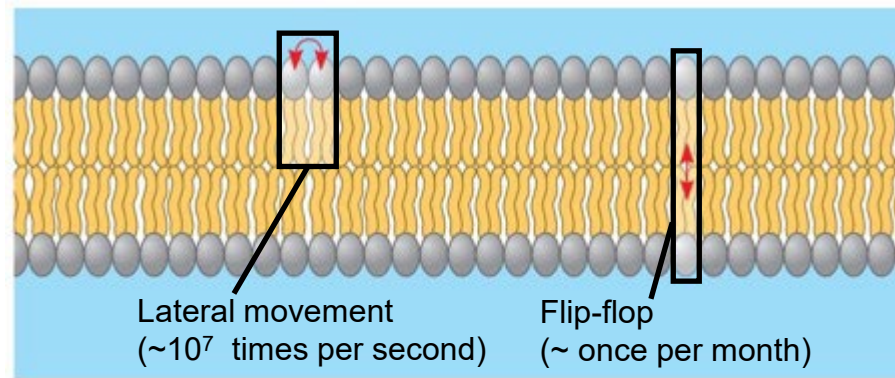
The **fluid mosaic model** - membrane is fluid with a “mosaic” of proteins embedded in it



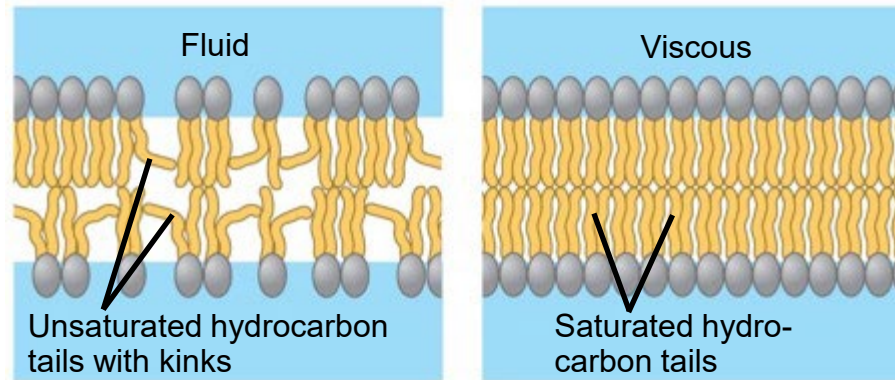
Campbell Biology,
12th ed, Fig 7.3

Amphipathic phospholipids

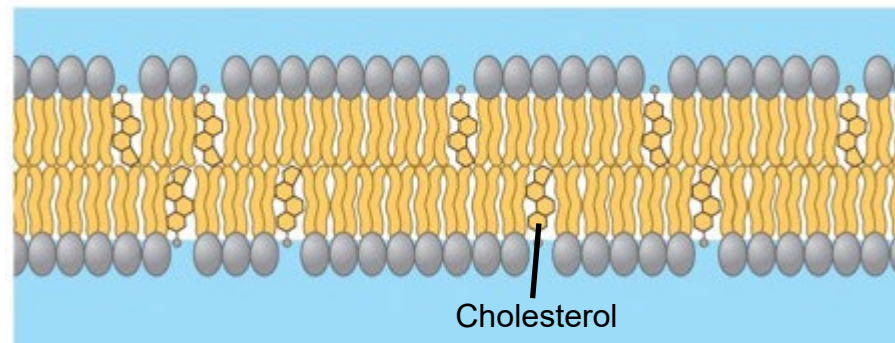




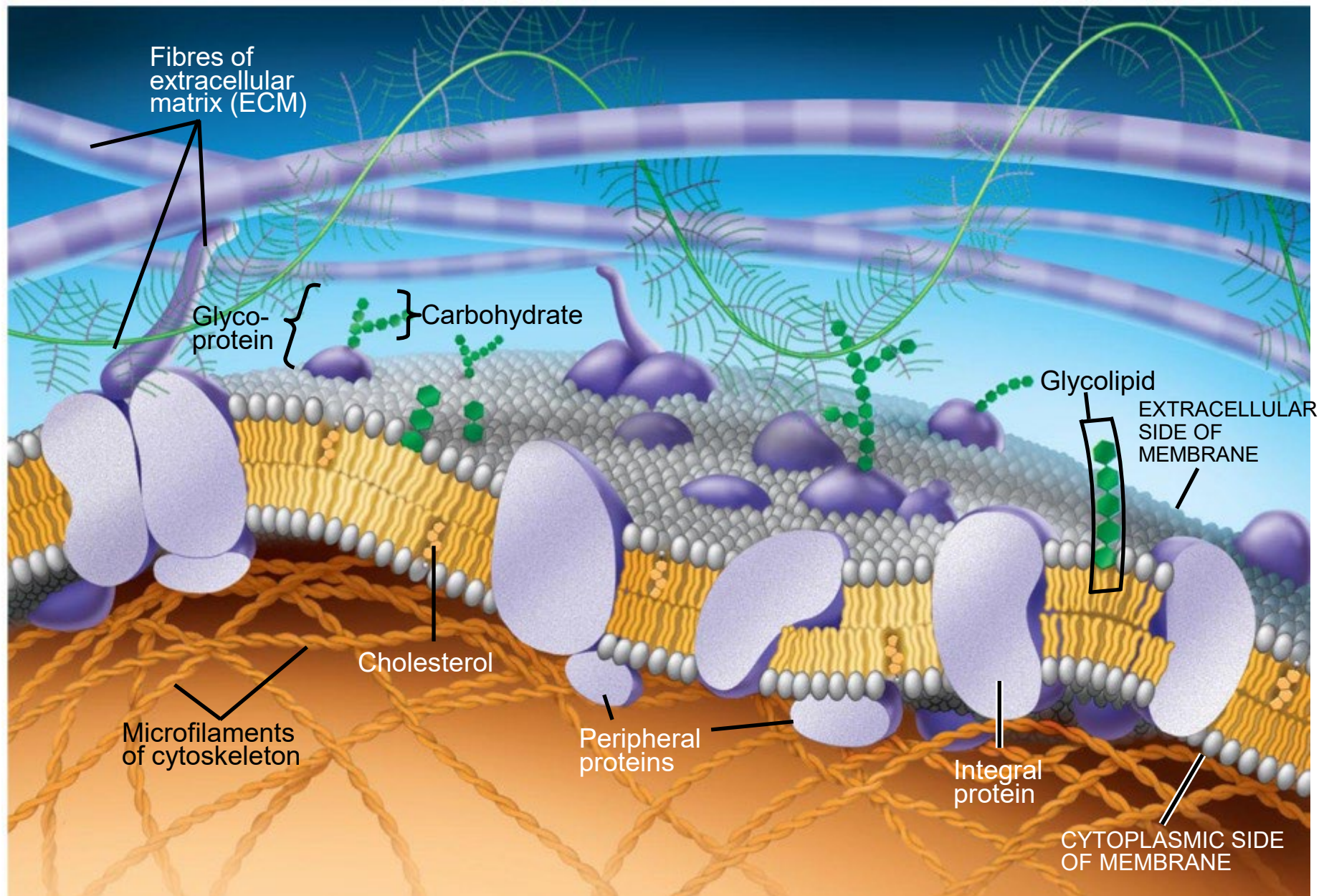
(a) Movement of phospholipids



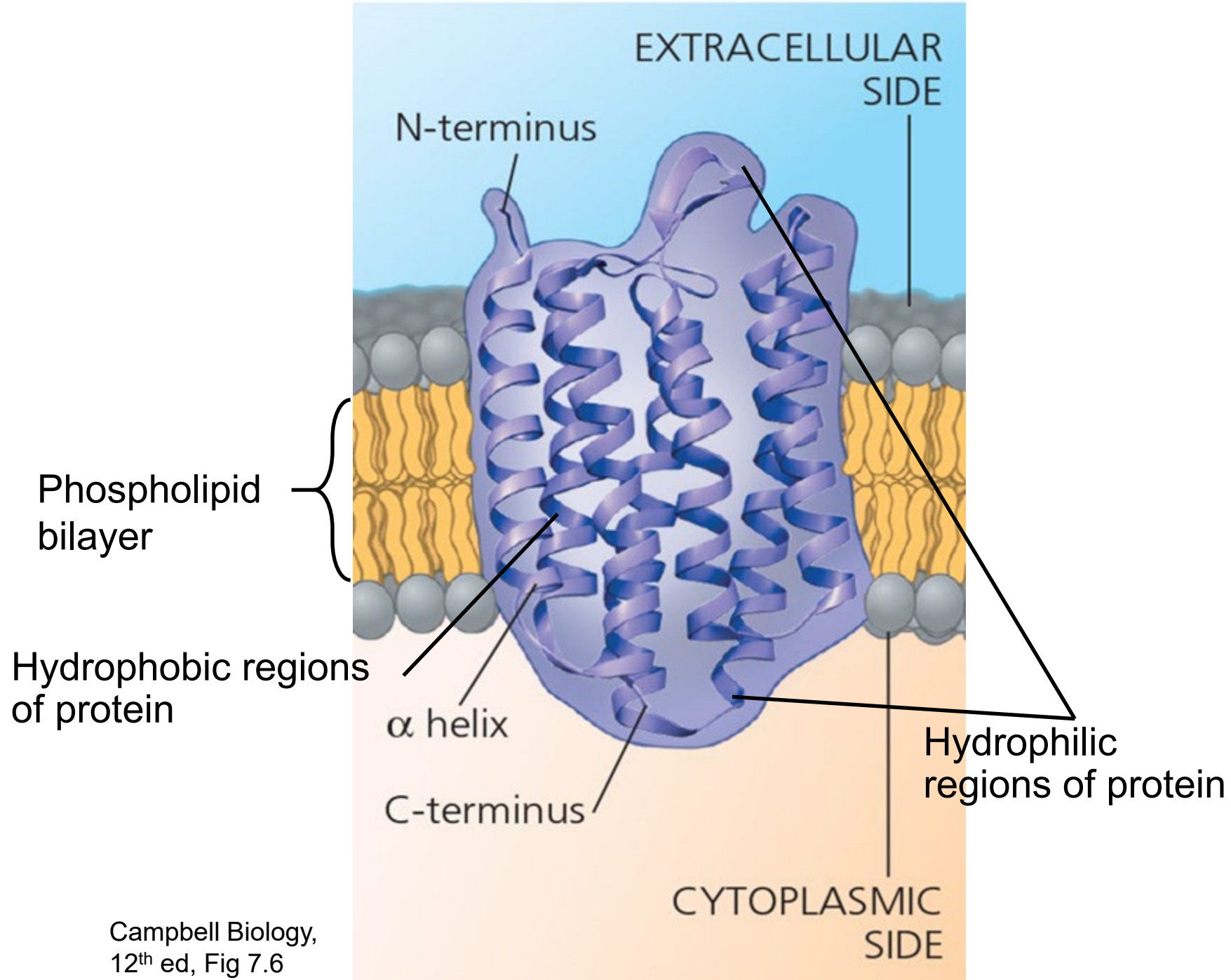
(b) Membrane fluidity



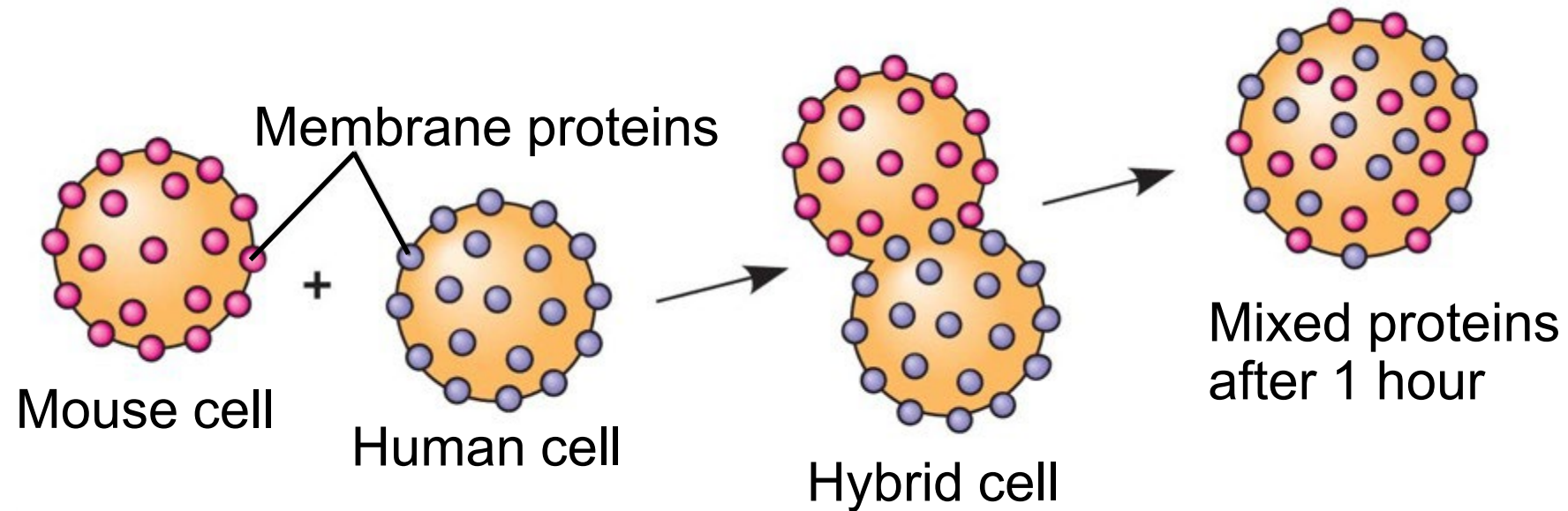
(c) Cholesterol within the animal cell membrane



Campbell Biology, 12th ed, Fig 7.3

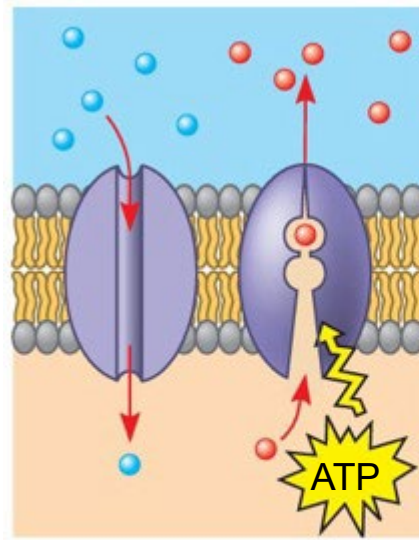


How do we know it is a fluid mosaic?

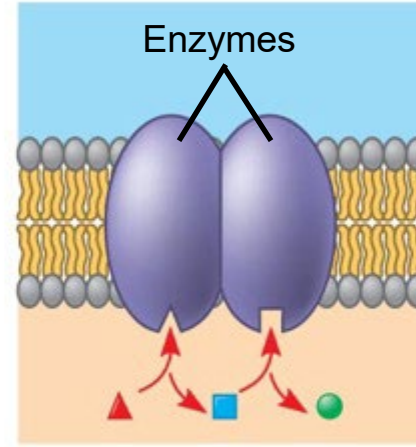


Campbell Biology, 12th ed, Fig 7.4

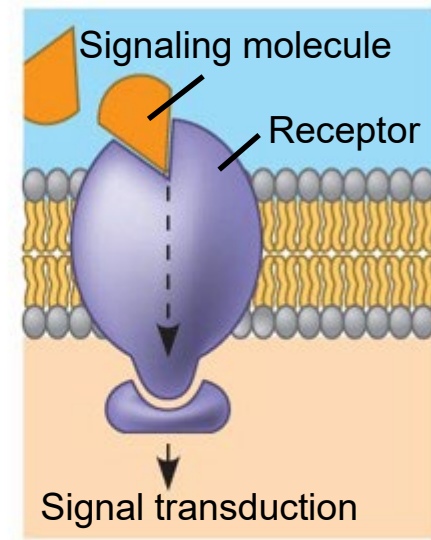
Six major functions of membrane proteins



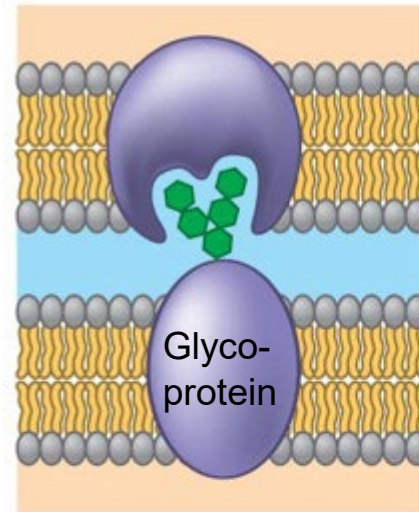
(a) Transport



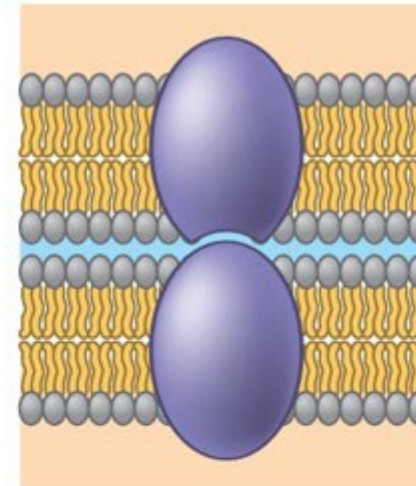
(b) Enzymatic activity



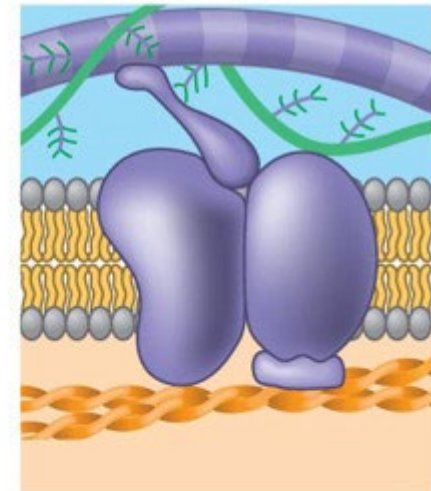
(c) Signal transduction



(d) Cell-cell recognition



(e) Intercellular joining



(f) Attachment to the cytoskeleton and extracellular matrix (ECM)

Concept 7.2: Membrane structure results in selective permeability

Cell membranes are lipid bilayers, which are impermeable to most essential molecules and ions

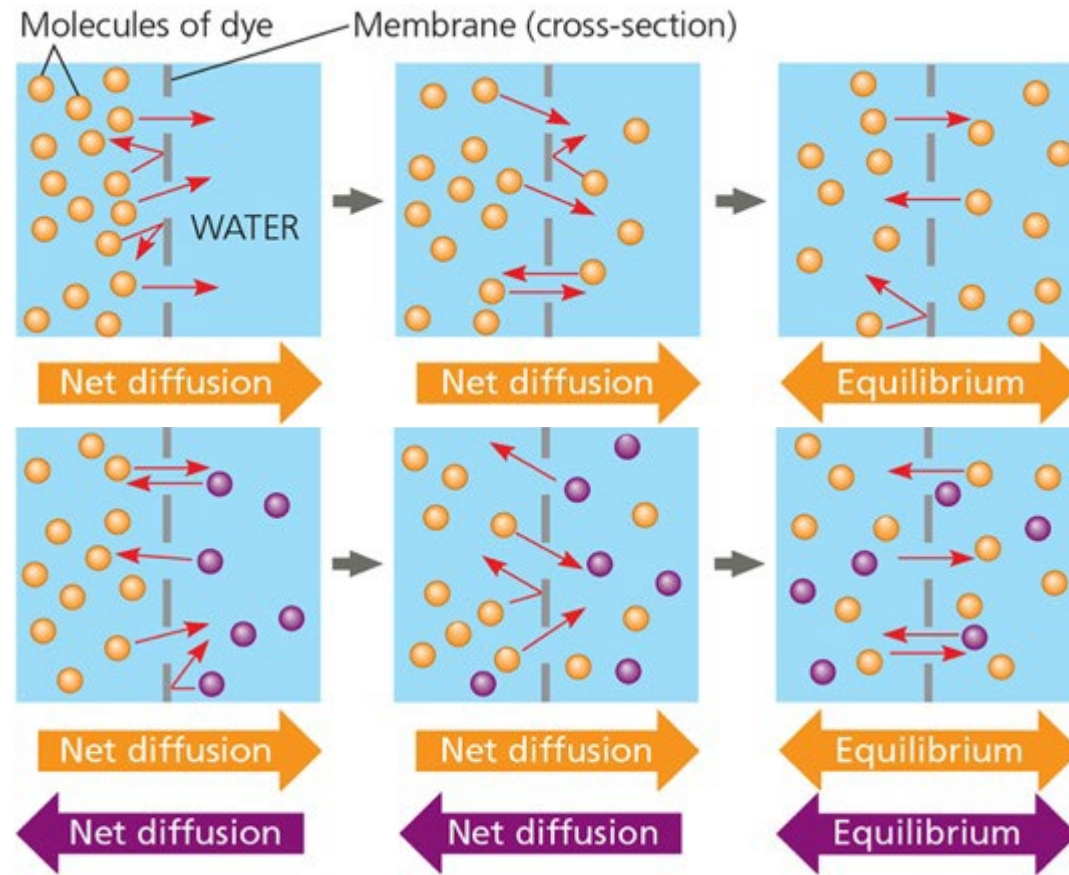
Some permeability to water molecules and a few other small, uncharged, molecules like oxygen and carbon dioxide.

Lipid bilayers are NOT permeable to:

- **ions** such as K^+ , Na^+ , Ca^{2+} , Cl^- , HCO_3^-
- small **hydrophilic molecules** like glucose and fructose
- **macromolecules** like proteins and RNA

Concept 7.3: Passive transport is diffusion of a substance across a membrane with no energy investment

DIFFUSION: with time – due to *random* motion molecules become *equally* distributed
i.e. to eliminate concentration gradients – provided molecules can cross the membrane



- Concentration gradient
- No work
- Diffusion across a biological membrane is passive transport (no ATP required)

Campbell Biology, 12th ed, Fig 7.11

Osmosis

Diffusion of water through a **selectively permeable membrane** into another aqueous compartment containing non-permeant solute at a higher concentration

Remember: plasma membrane is semi-permeable!

Why is osmosis important?

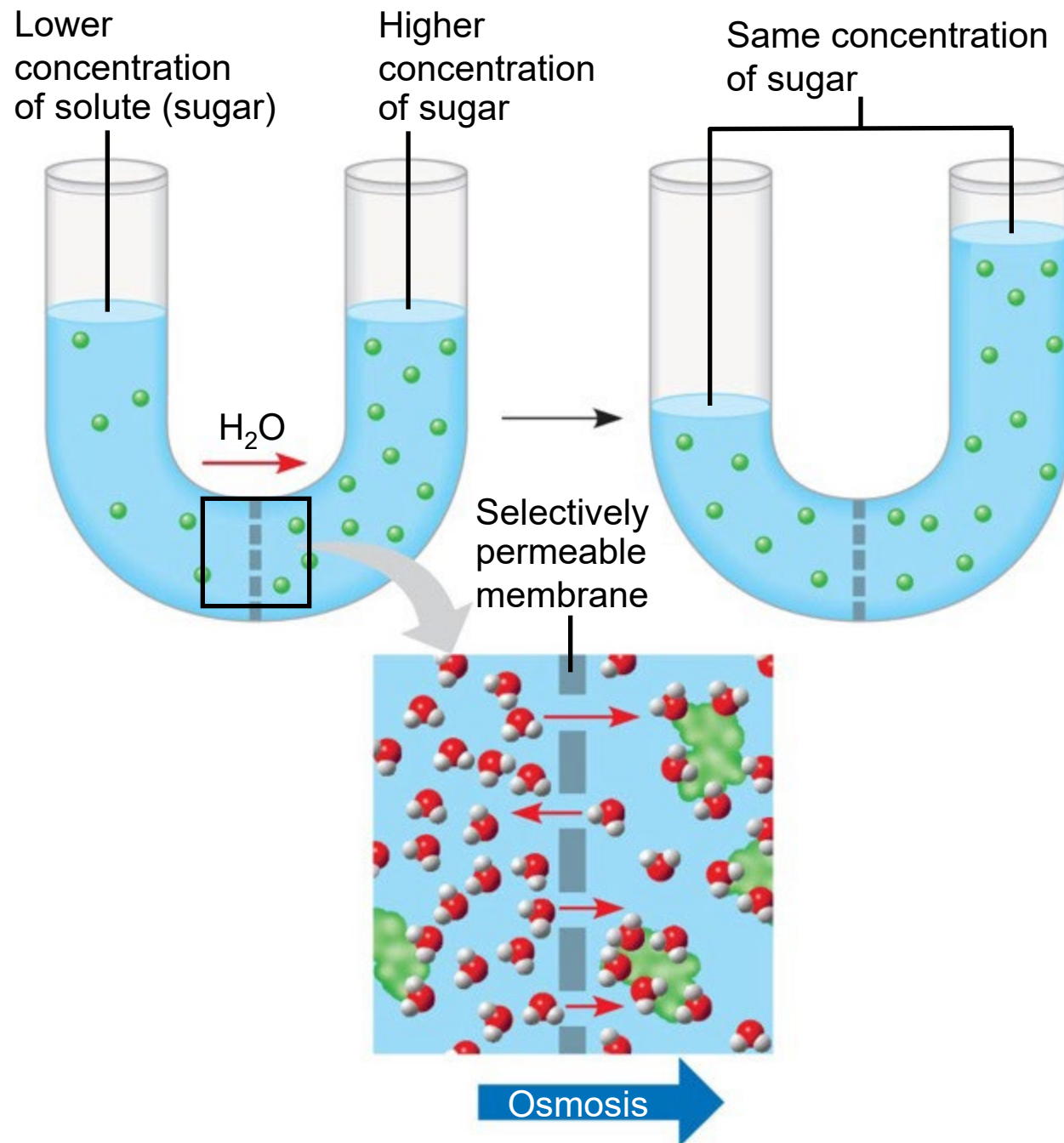
Body Fluid Compartments (70 kg male)

	% of Body Weight	% of Total Body Water	Volume (Litres)
ECF	27	45	19
<i>Plasma</i>	<i>4.5</i>	<i>7.5</i>	<i>3.2</i>
<i>ISF</i>	<i>12.0</i>	<i>20.0</i>	<i>8.4</i>
<i>Dense CT water</i>	<i>4.5</i>	<i>7.5</i>	<i>3.2</i>
<i>Bone water</i>	<i>4.5</i>	<i>7.5</i>	<i>3.2</i>
<i>Transcellular</i>	<i>1.5</i>	<i>2.5</i>	<i>1.0</i>
ICF	33	55	23
TBW	60%	100%	42

What are “osmotica”?

(solutes that are osmotically active)

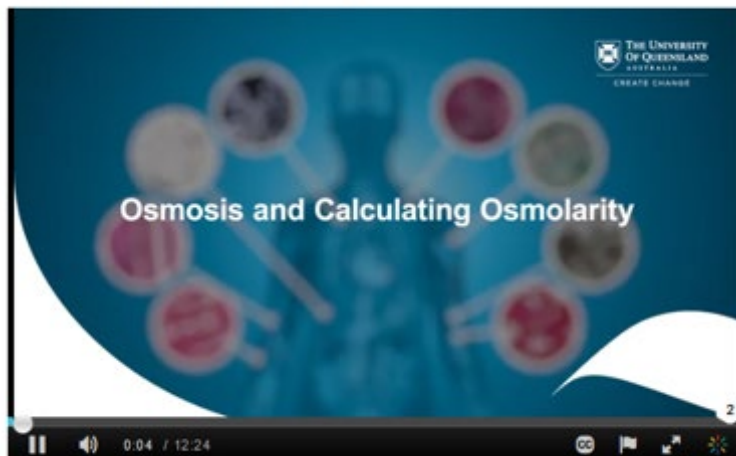
- Ions (Na^+ , K^+ , Cl^- , etc)
- Sugars
- Proteins
- (Nutrients!!)



Videos to explain:

1. Osmosis and Calculating Osmolarity
2. Osmosis and Tonicity

Osmosis and Calculating Osmolarity

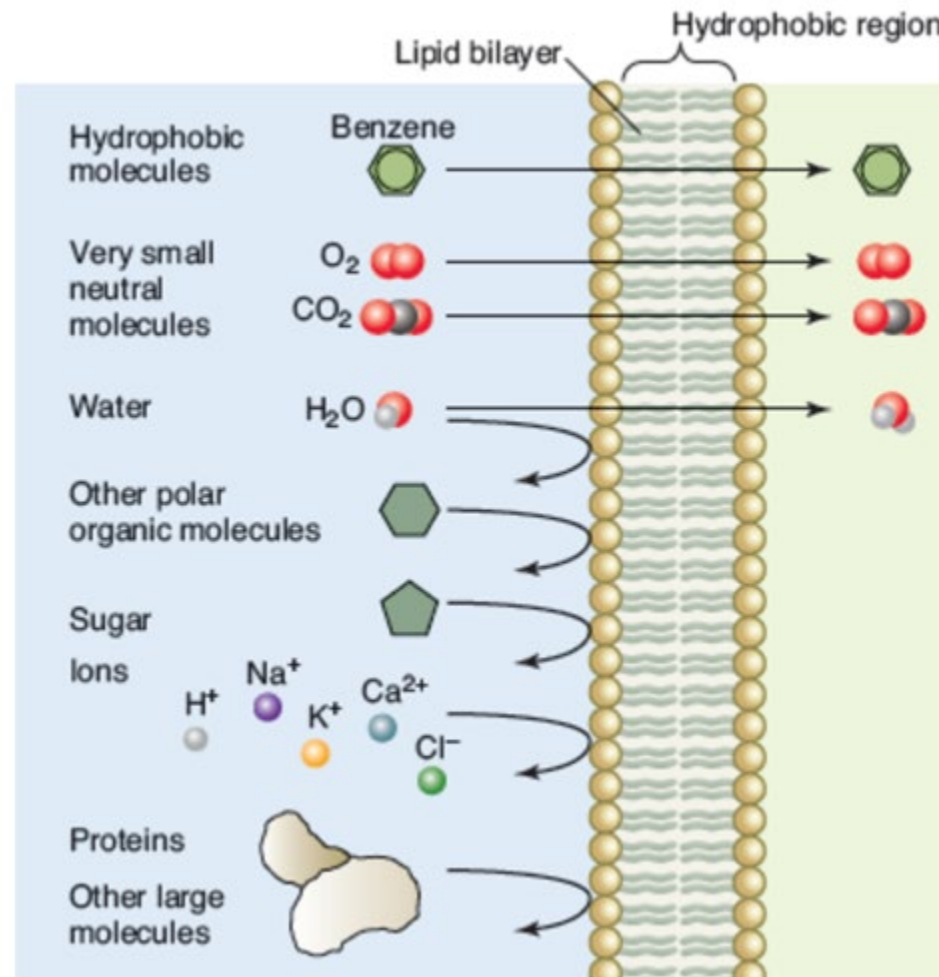


Osmosis and Tonicity

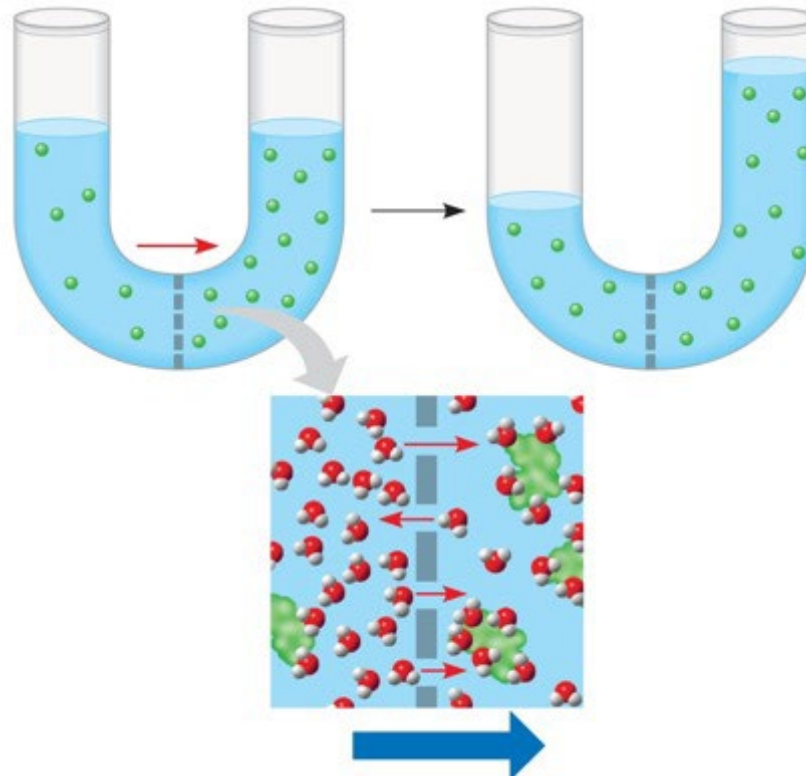


You will need the content of these videos for Workshop 2!

Selective permeability of the lipid bilayer cell membrane

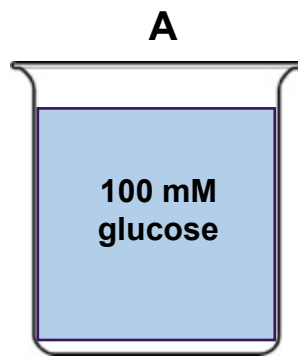


Osmosis – diffusion of water from low to high solute concentration across a selectively permeable membrane

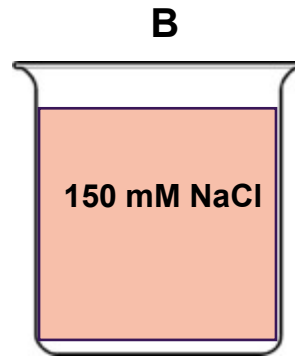


1. Does the solute cross the cell membrane by simple passive diffusion?
2. If the solute dissolves, does it remain intact or dissociate into ions?

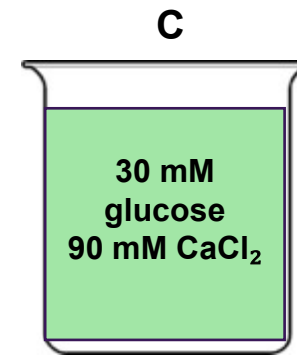
$$\text{Osmolarity (Osm/L)} =$$
$$\text{Molar concentration} \times \text{number of molecules or ions in solution}$$



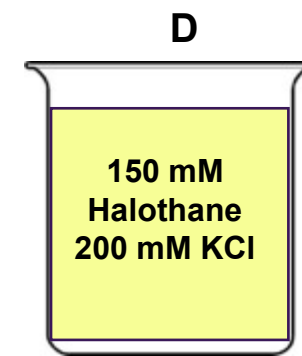
100 mOsm/L



300 mOsm/L



300 mOsm/L



400 mOsm/L

Tonicity : The ability of a solution to cause a cell to gain or lose water

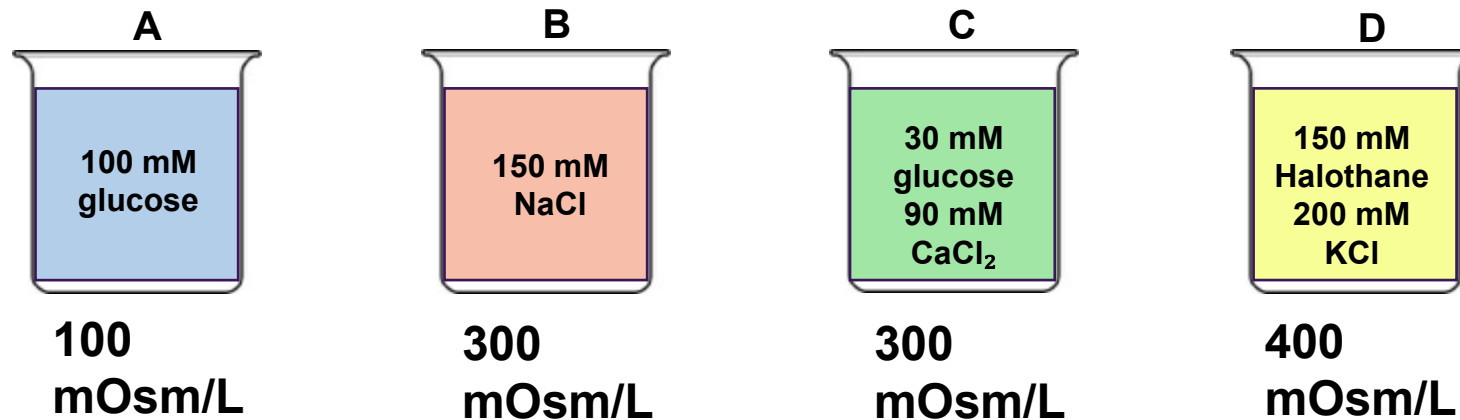
Isotonic solution: Solute concentration is the same as that inside the cell;
no net water movement across the plasma membrane

Hypertonic solution: Solute concentration is greater than that inside the cell;
cell loses water

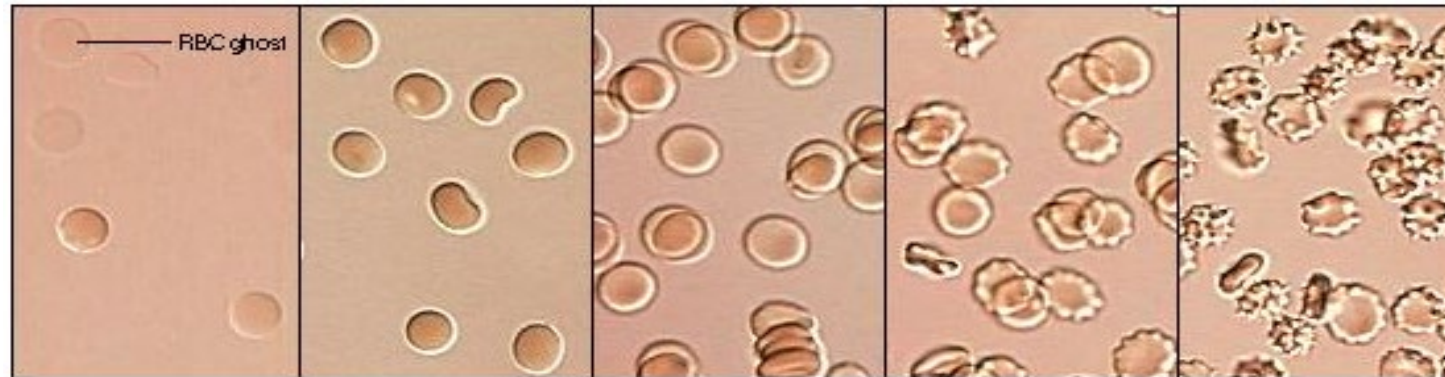
Hypotonic solution: Solute concentration is less than that inside the cell;
cell gains water

Based on the osmolarity of each solution,

- (i) compare the tonicity with the contents of red blood cells (approx. 300 mOsm/L)
- (ii) draw diagrams to illustrate what would happen to red blood cells if placed in the solution



Effect of tonicity on human red blood cells

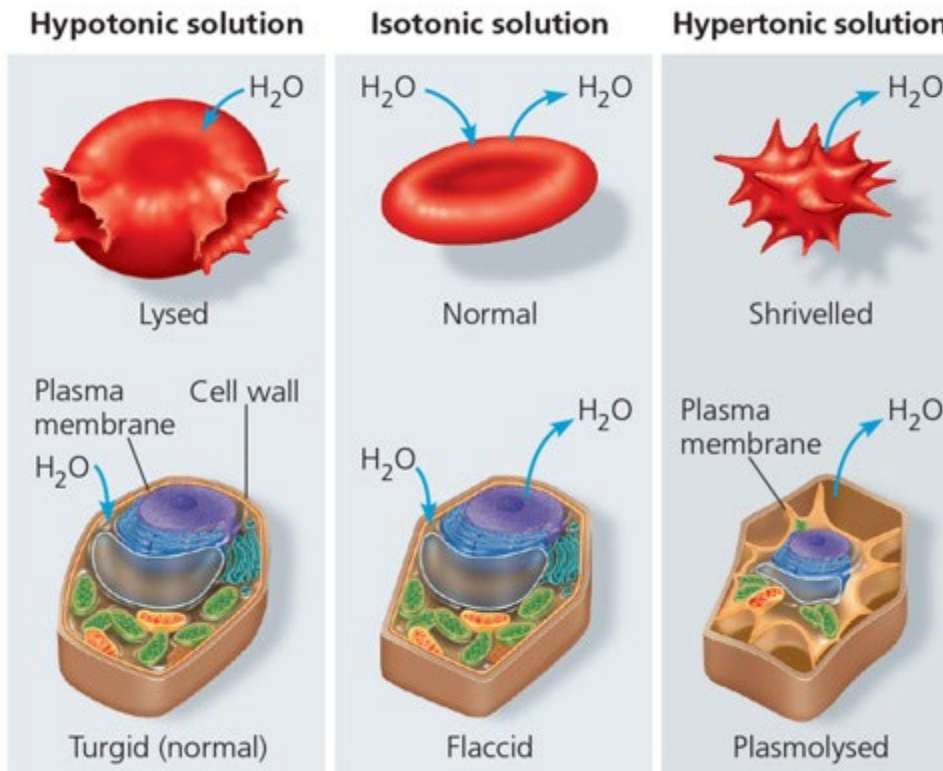


50 mOsm/L		300 mOsm/L		500 mOsm/L
Hypotonic	← Isotonic →			Hypertonic
Cells lysed	Cell volume increases	Normal cell volume	Cell volume decreases	Cell volume decreases more

Osmosis – animal vs. plant cells

(a) Animal cell. An animal cell fares best in an isotonic environment unless it has special adaptations that offset the osmotic uptake or loss of water.

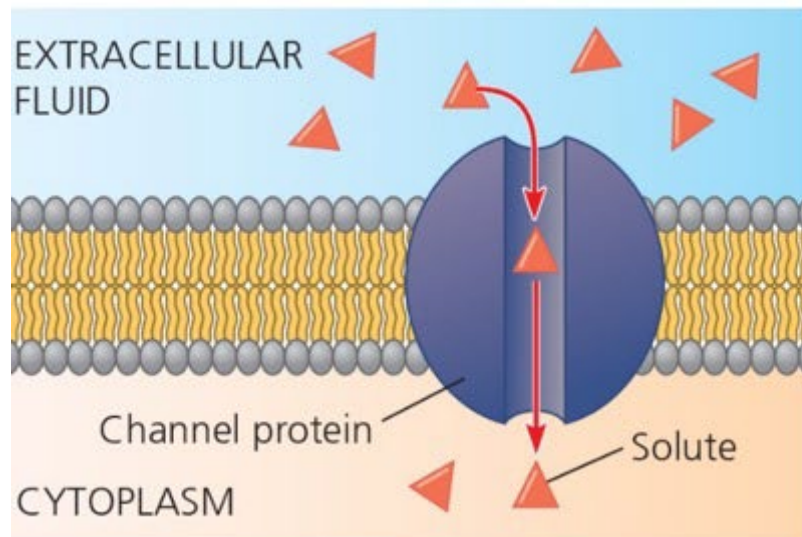
(b) Plant cell. Plant cells are turgid (firm) and generally healthiest in a hypotonic environment, where the uptake of water is eventually balanced by the wall pushing back on the cell.



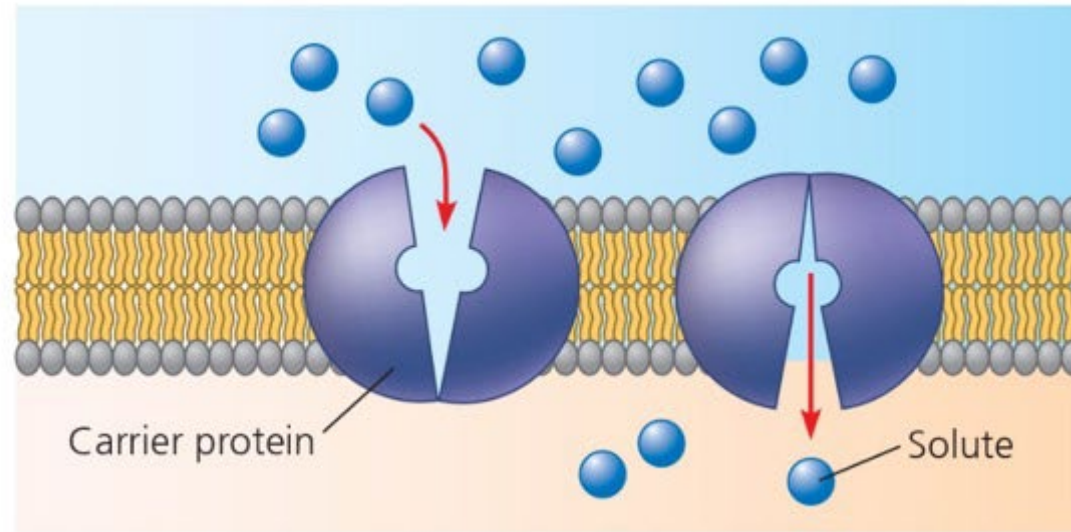
Facilitated Diffusion: Passive Transport Aided by Proteins

In **facilitated diffusion**, transport proteins speed up the passive movement of molecules across the plasma membrane

2 types of transport proteins for facilitated diffusion:



(a) A channel protein



(b) A carrier (or transporter) protein

Transport Proteins

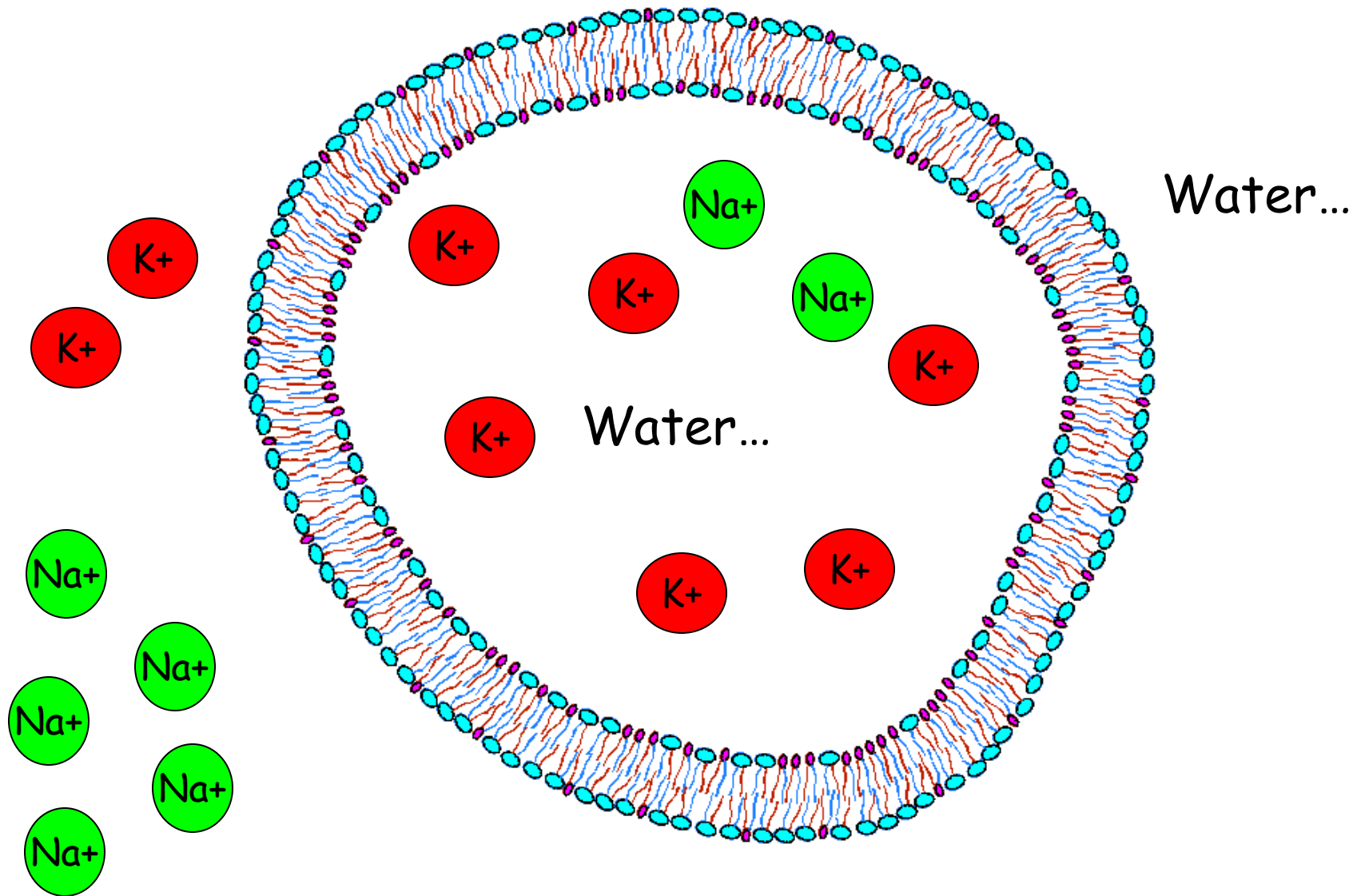
Channels and transporters (or carriers) are proteins that allow passage of hydrophilic substances across the membrane

Facilitated diffusion occurs along the solute concentration gradient and does NOT require energy

Transport proteins include

- **Aquaporins** and **ion channels** that open or close in response to a stimulus (**gated channels**)
- **Carriers (or transporters)** that bind to molecules and change shape to shuttle them across membrane – e.g. glucose transporters; amino acid transporters

Getting across the membrane

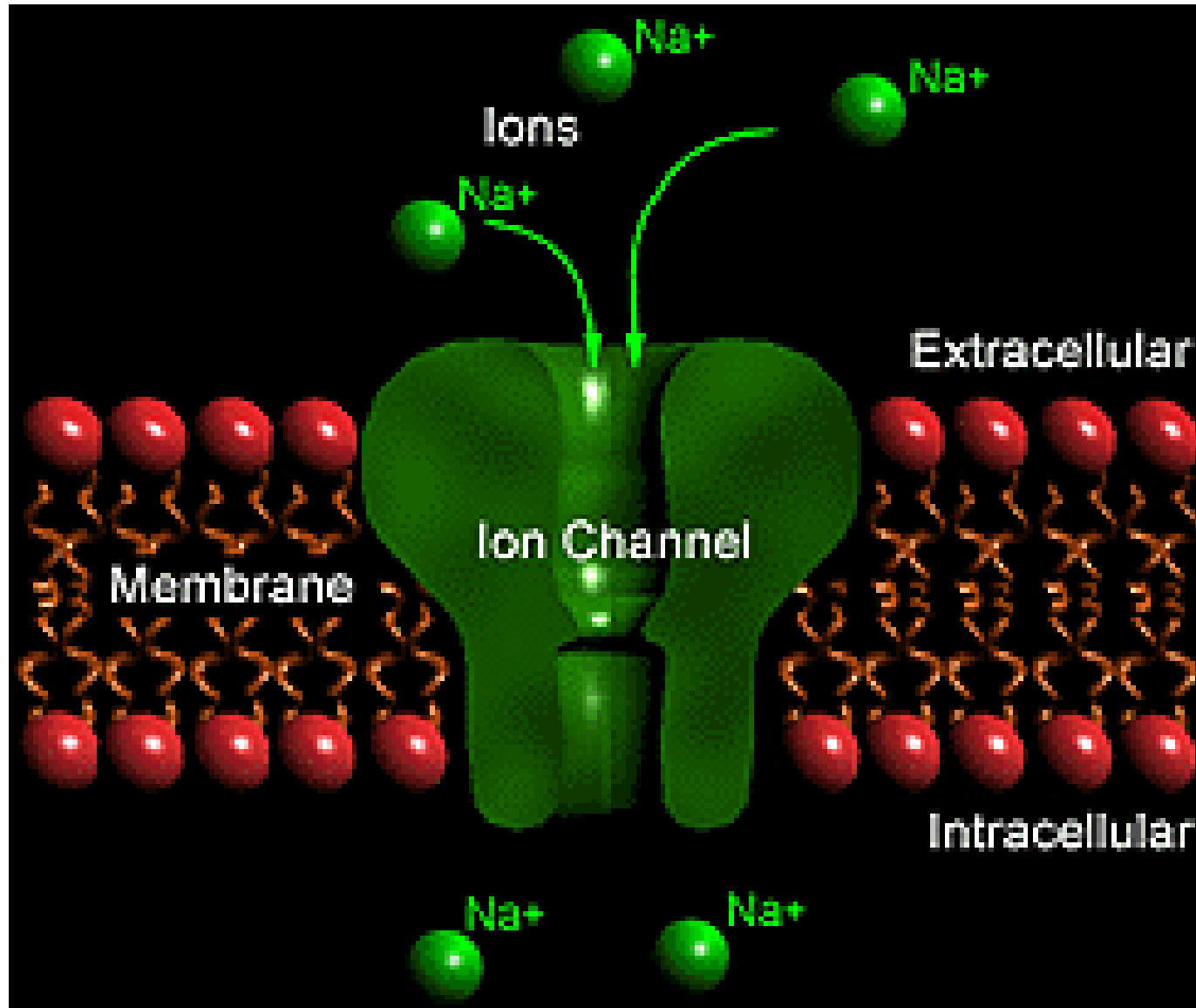


Membrane is a diffusion barrier for most molecules!!!!

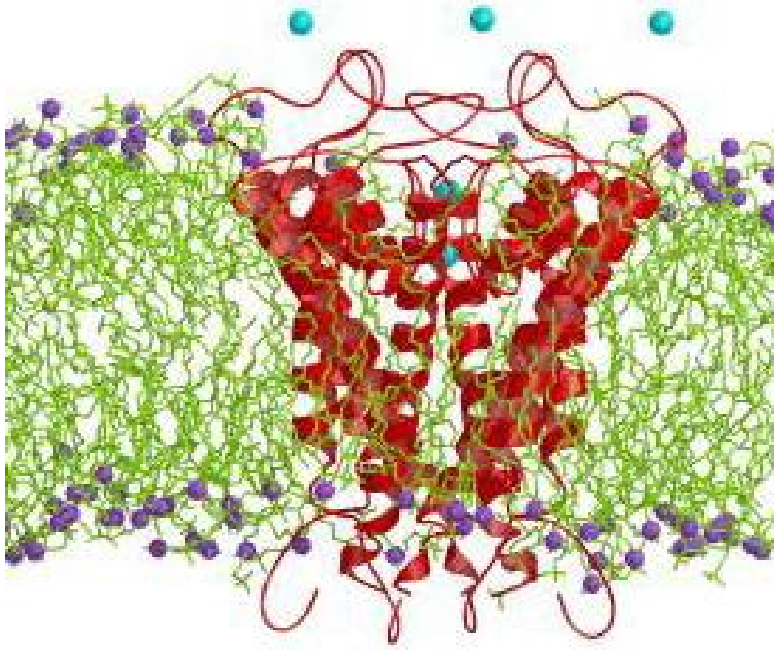
Site	Laboratory test	Reference range
Blood/serum	Sodium (mmol/L)	135-149
	Potassium (mmol/L)	3.5-4.7
	Urea (mmol/L)	3.3-7.0
	Creatinine (μ mol/L)	60-120
	Chloride (mmol/L)	96-106
	Calcium (mmol/L)	2.2-2.6
	Phosphate (mmol/L)	0.8-1.4
	Magnesium (mmol/L)	0.7-1.1
	Albumin (g/L)	35-55
	Glucose (mmol/L)	3.6-6.1
	Creatine phosphokinase (U/L)	42-245
	Arterial pH	7.35-7.45
	PCO ₂ (mm Hg)	35-45
	PO ₂ (mm Hg)	83-105
	Bicarbonate (mmol/L)	22-26
	Anion gap (mEq/L) [mmol/L]	5-14

Na⁺ = 140 mM K⁺ = 4.5 mM

These transmembrane proteins form “pores” through which ions and molecules can pass

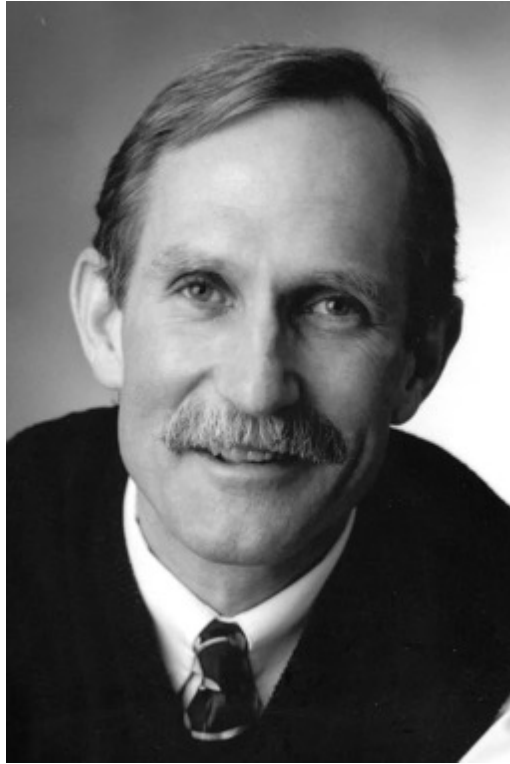


Potassium Ion Channel Structure

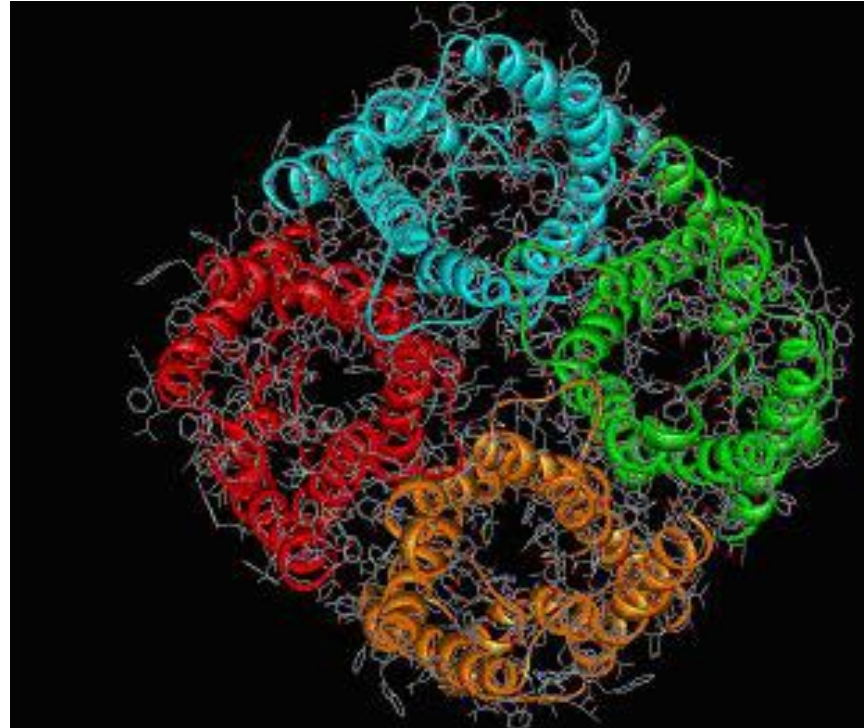


Rod MacKinnon
Nobel Prize for Chemistry
2003

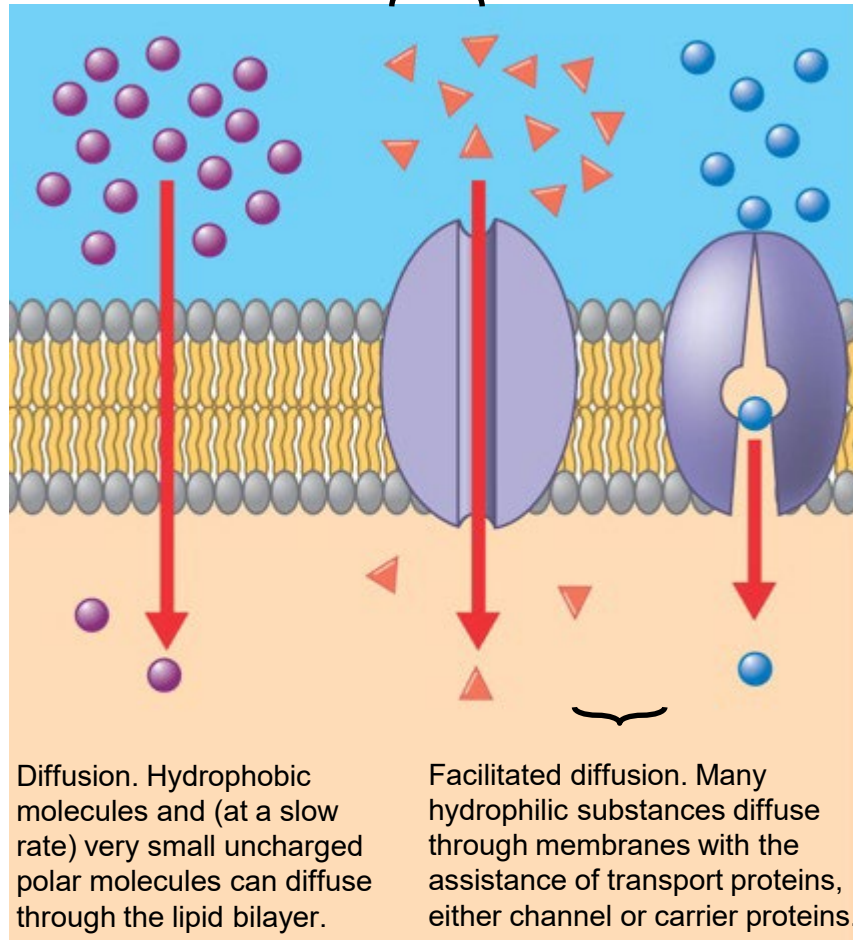
The water channel: Aquaporin



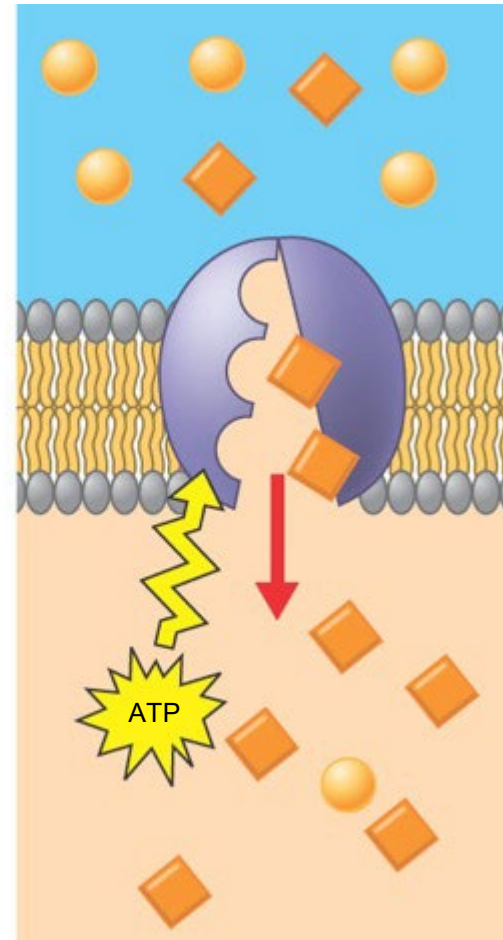
Peter Agre
Nobel Prize for
Chemistry 2003



Passive



Active



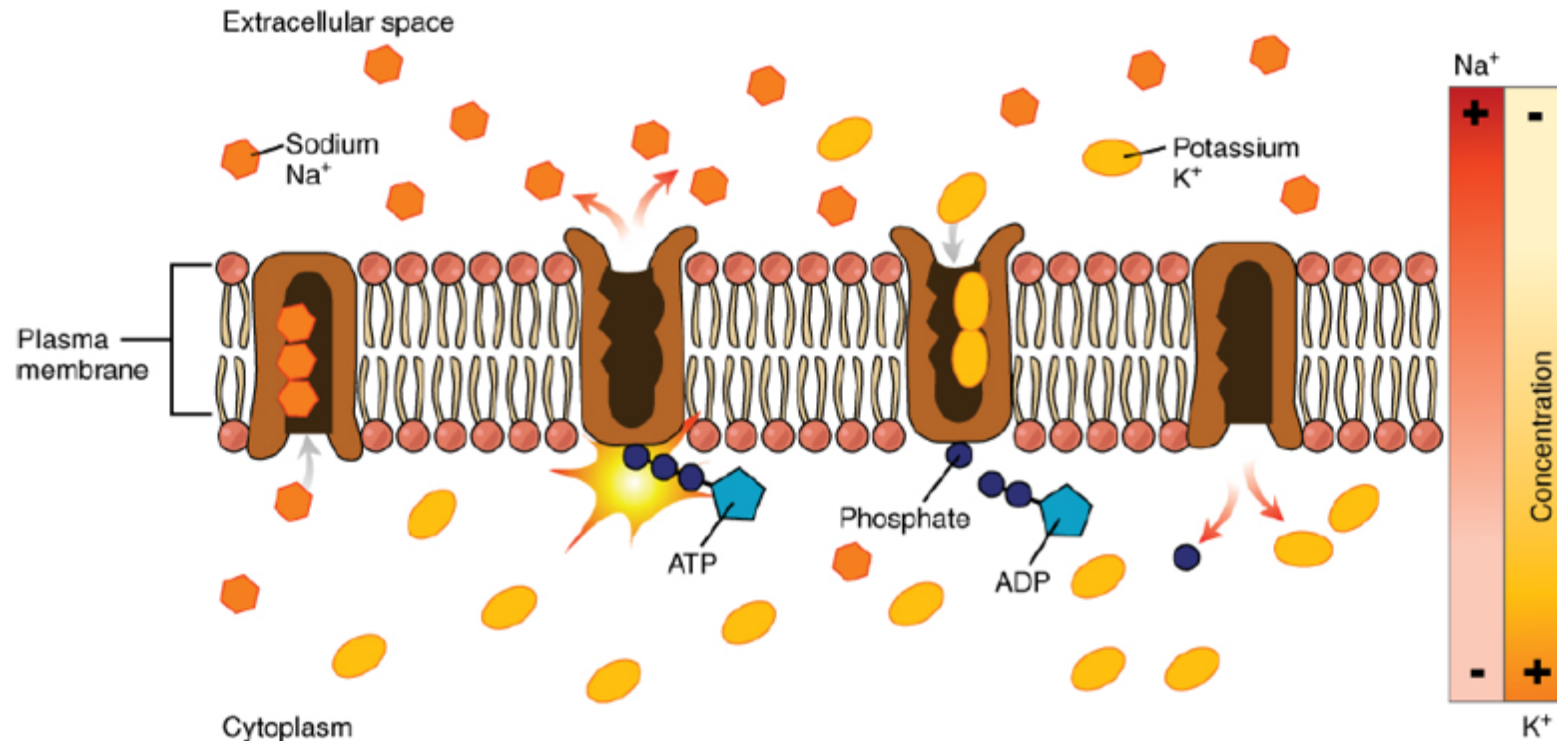
Passive = DOWN a concentration gradient

Concept 7.4: Active transport uses energy to move solutes against their gradients

- Only transporter (or carrier) proteins can mediate active transport of hydrophilic substances across the membrane; NOT channels
- Transport solutes **against concentration gradient**
- Requires work – from energy of ATP
- e.g. Na^+/K^+ -ATPase = sodium-potassium pump

An **electrogenic pump** is a transporter that generates voltage across a membrane

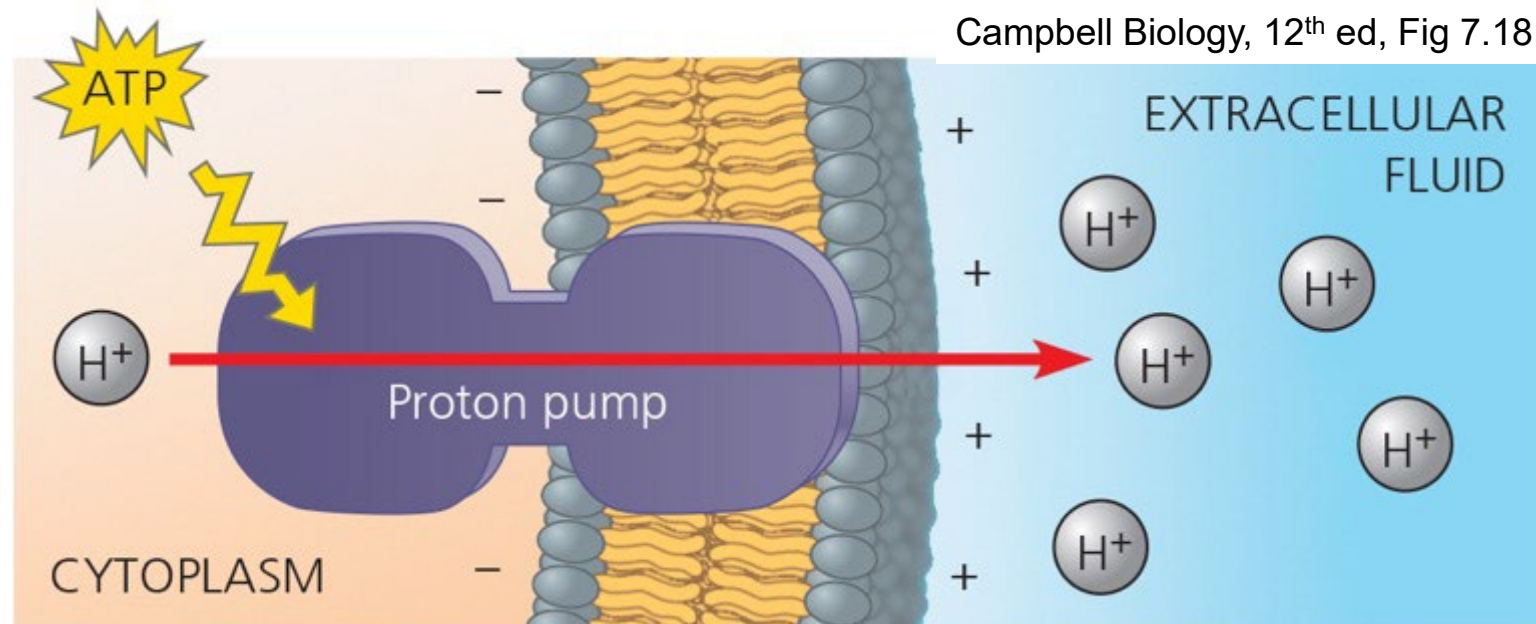
The **sodium-potassium pump** is the major electrogenic pump of animal cells – see Nervous Systems Module



An **electrogenic pump** is a transporter that generates voltage across a membrane

The **sodium-potassium pump** is the major electrogenic pump of animal cells – see Nervous Systems Module

The main electrogenic pump of plants, fungi, and bacteria is a **proton pump**



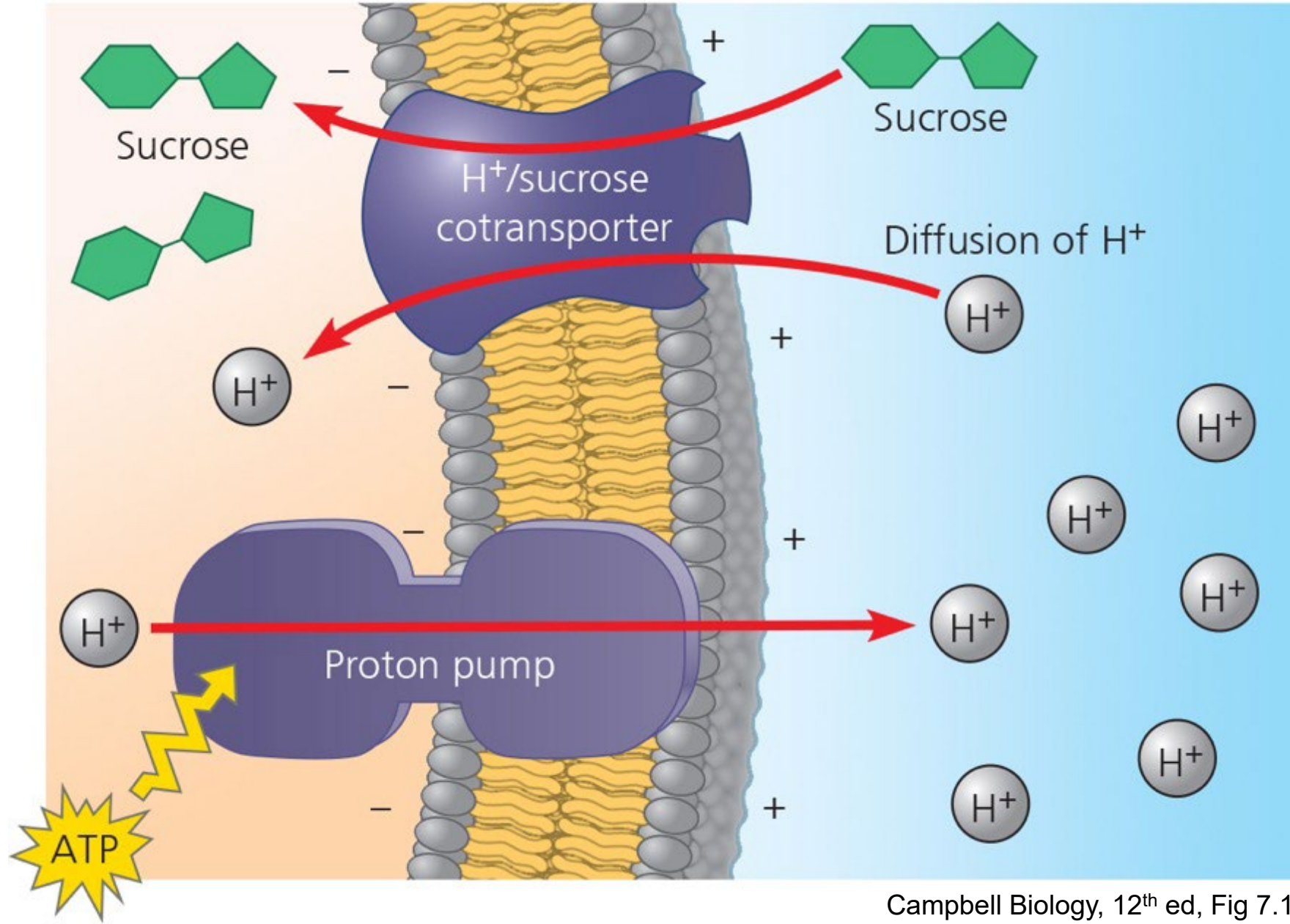
Cotransport: Coupled Transport by a Membrane Protein

Cotransport occurs when active transport of a solute indirectly drives transport of another solute

Plants commonly use the gradient of hydrogen ions generated by proton pumps to drive active transport of nutrients into the cell

Animal cells: Na^+/K^+ -ATPase working together with, for example, the Na^+ -glucose cotransporter to drive glucose uptake against its concentration gradient

Cotransport: active transport driven by a concentration gradient



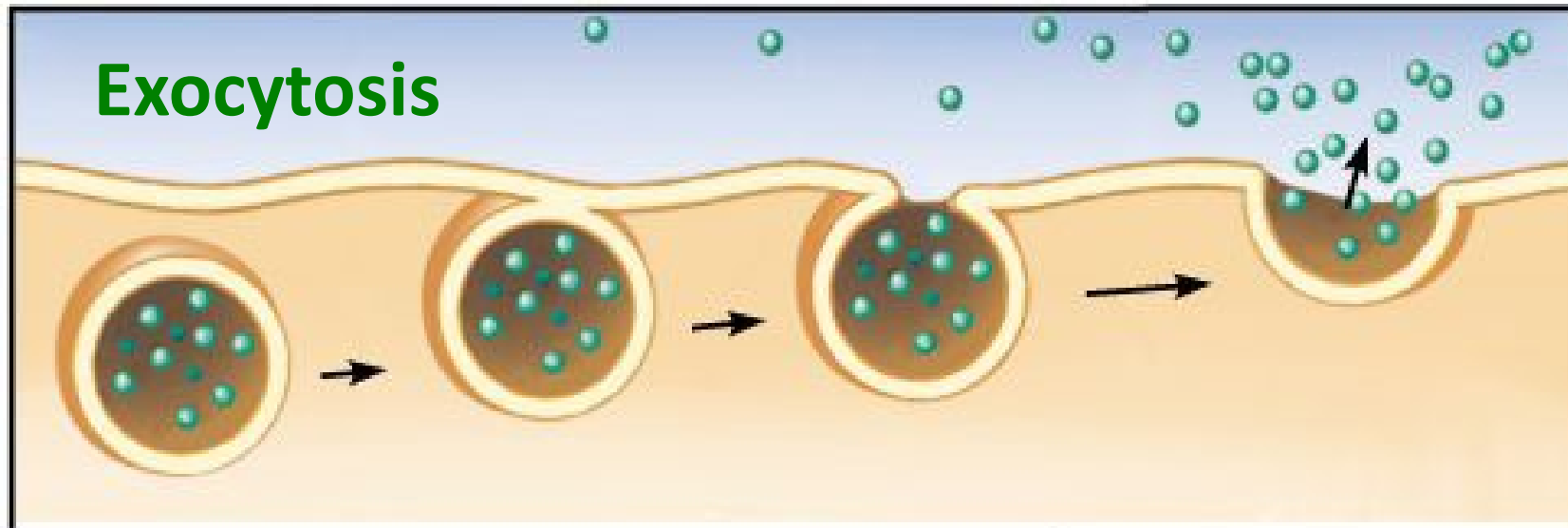
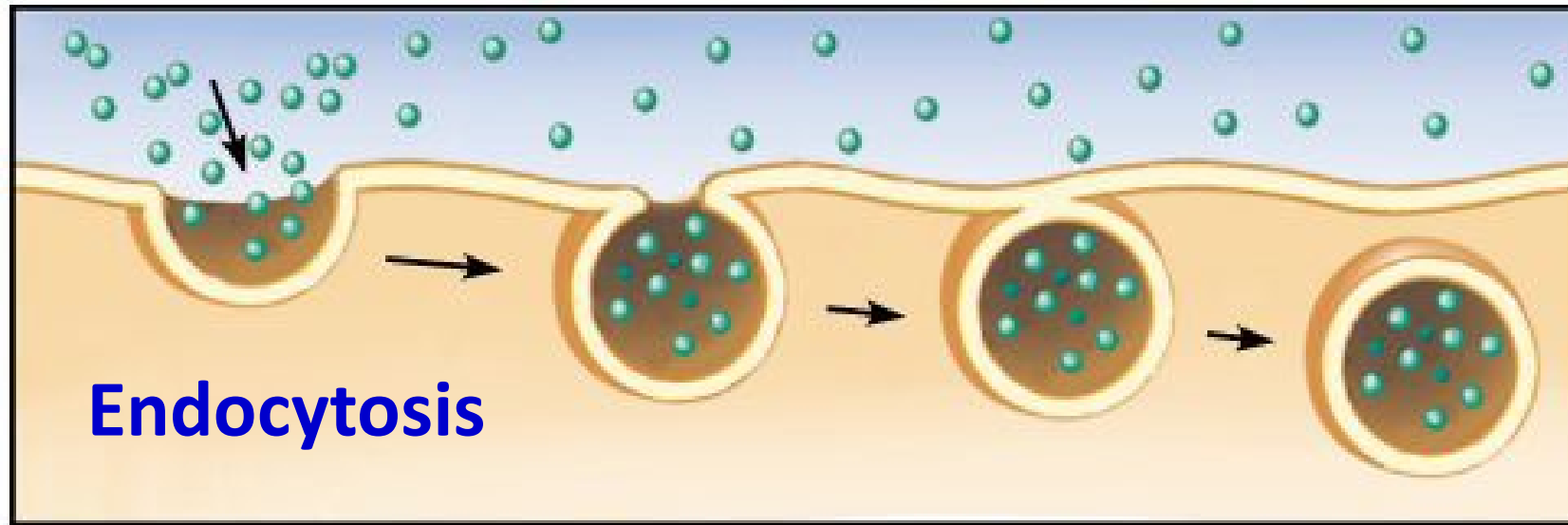
Another important video for you to study:
Membrane Transport – Active Transport and Co-Transport

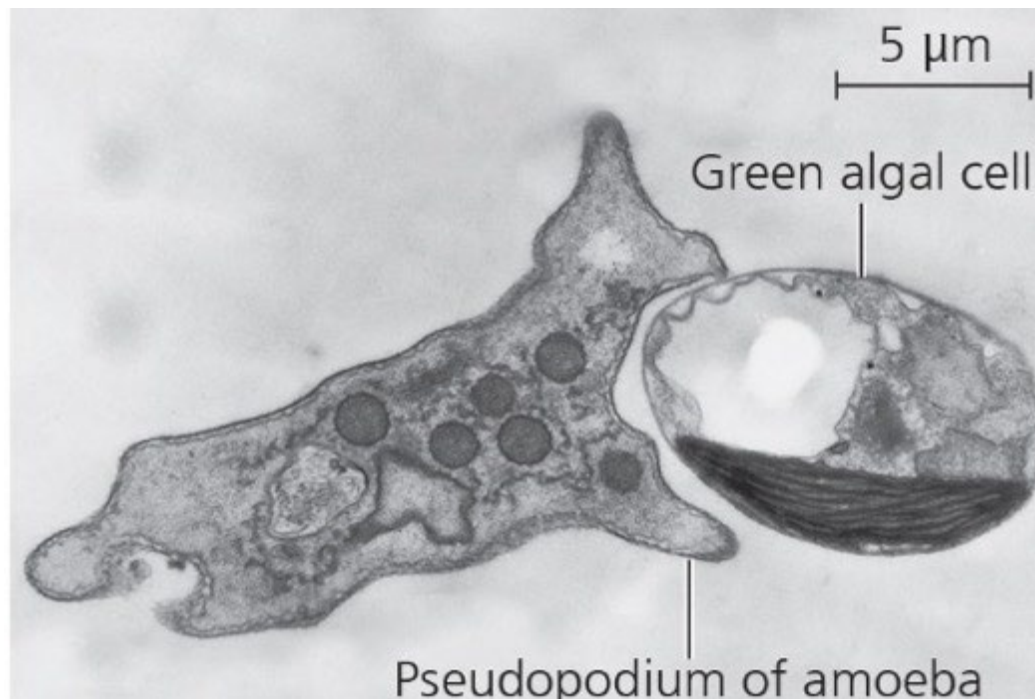
Membrane Transport - Active Transport and Co-Transport



The content of this video is also needed for Workshop 2!

Concept 7.5: Bulk transport across the plasma membrane occurs by exocytosis and endocytosis



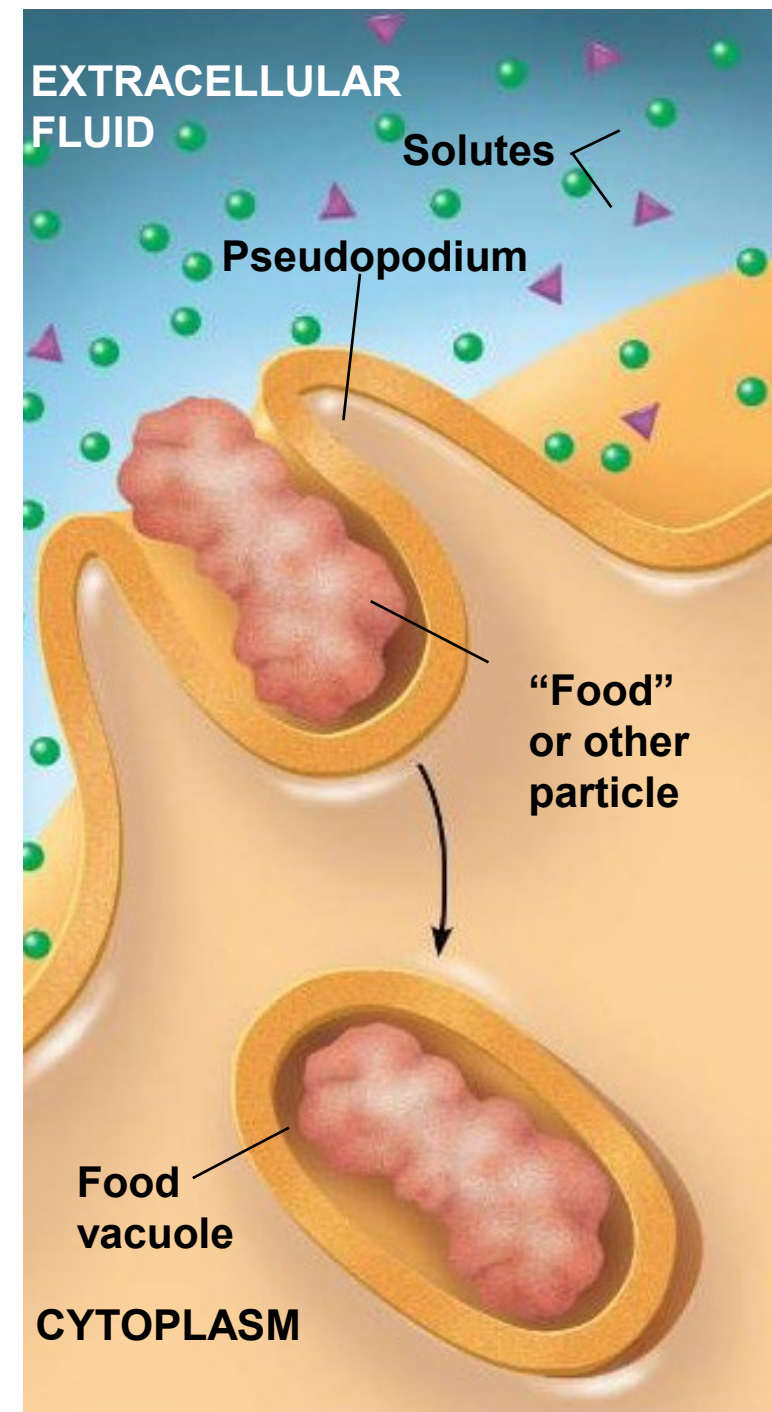


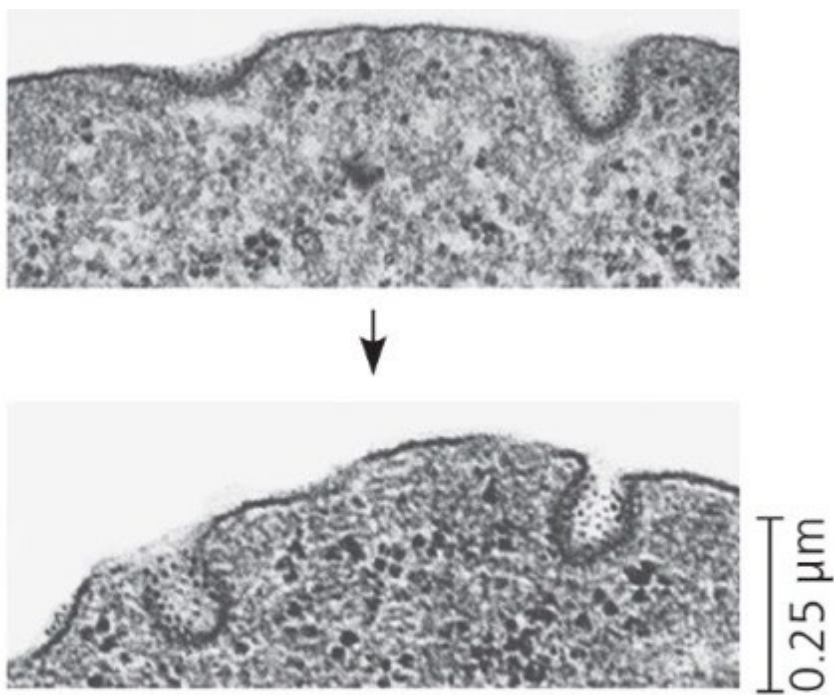
An amoeba engulfing a green algal cell via phagocytosis (TEM).

Types of Endocytosis

Phagocytosis – engulfing particles

Campbell Biology, 12th ed, Fig 7.21



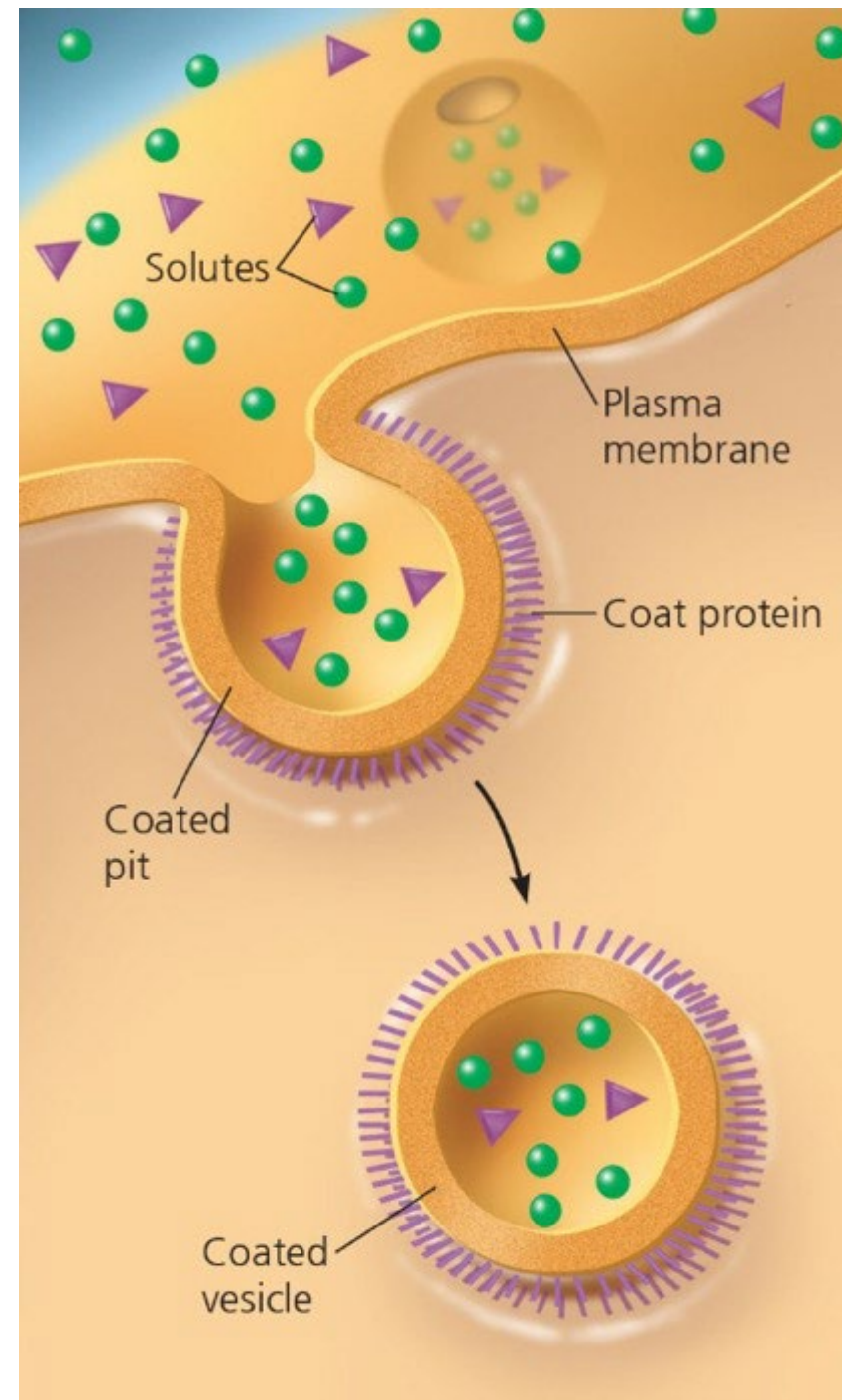


Pinocytotic vesicles forming (TEMs).

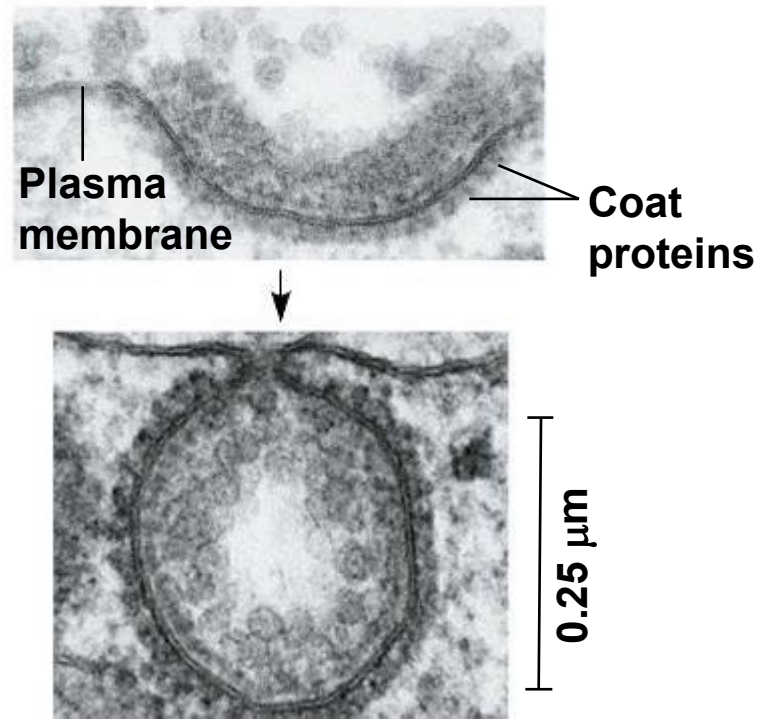
Types of Endocytosis

Pinocytosis – cells drinking!

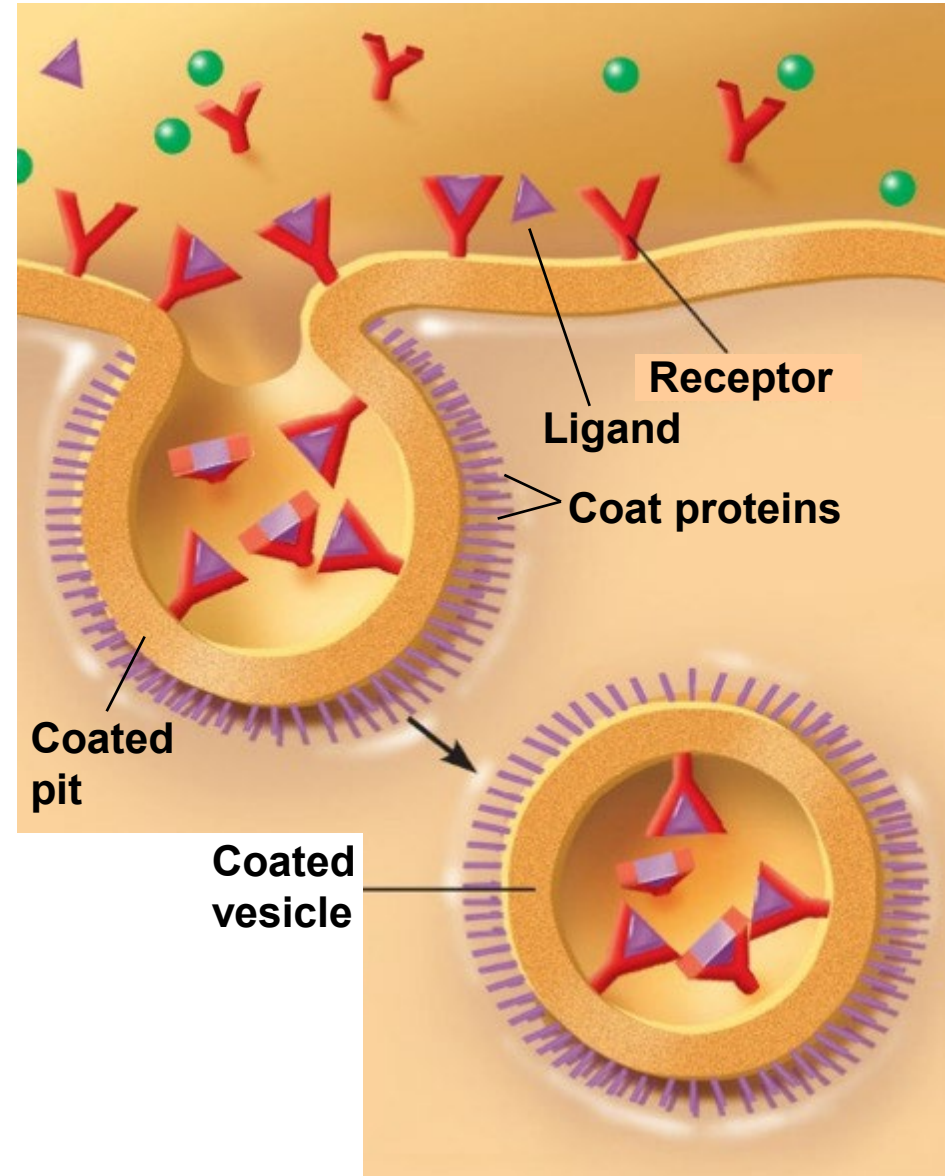
Campbell Biology, 12th ed, Fig 7.21



Types of Endocytosis – Receptor-mediated endocytosis



Top: A coated pit. **Bottom:** A coated vesicle forming during receptor-mediated endocytosis (TEMs).



Exocytosis

Things packaged up in the cells to get out!

Where does this happen in the body??

